MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION, NOVEMBER 2024
Discipline Specific Core Course - MG1DSCECT101 - COMPUTER FUNDAMENTALS AND BASICS OF PC HARDWARE
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A
Multiple Choice Questions
Answer all questions. Each question carries 1 mark

1.	Bit	[U]	[1]
2.	Arithmetic Logic Unit (ALU)	[U]	[1]
3.	Application development	[U]	[1]
4.	Raw facts and figures	[K]	[1]
5.	To manage the execution of instructions and coordinate the activities of the CPU	[U]	[2]
6.	Random Access Memory(RAM)	[U]	[2]
7.	To connect a monitor or Projector for video output	[U]	[2]
8.	Virtualizing data storage	[An]	[3]
9.	Artificial Intelligence	[An]	[3]
10.	L1 is faster and smaller than L3	[An]	[3]

(10 x 1 = 10)

Part B
Short Answer Questions
Answer 4 questions. Each question carries 5 marks

11. Input stage, Processing stage (Fetch-Decode- Execute cycle), Output stage, Storage stage
    Explanation of input/output/storage stage - 1 mark each
    Explanation of Processing stage - 2 marks [U] [1]

12.

Primary Memory	Secondary Memory
Volatile (data is lost when power is off).	Non-volatile (data remains even when power is off).
Used for temporary storage of data and instructions during processing.	Used for permanent storage of data and programs.

Typically smaller in capacity (e.g., 4 GB - 32 GB).	Larger in capacity (e.g., 500 GB - several TB).
More expensive per GB.	Less expensive per GB.
Directly accessed by the CPU.	Accessed indirectly (requires I/O operations).
Handles data that is actively in use. Hence called the working memory of a computer.	Stores data for long-term use and retrieval.
RAM (Random Access Memory), Cache Memory.	HDD (Hard Disk Drive), SSD (Solid-State Drive), Optical Disks.

Any 4 relevant differences - 4 marks
Any example of each - 1 mark [U] [1]

13. Flash drives are portable, lightweight devices that offer various storage capacities, making them suitable for diverse data needs. They are durable due to the lack of moving parts, providing resistance to physical damage. Flash drives are user-friendly, featuring plug-and-play functionality that requires no additional software for access. They provide faster data transfer speeds compared to traditional USB storage devices. Their primary functions include data storage, easy data transfer between computers and devices, backup of important files, and file sharing.
    Definition - 1 mark
    Features - 2 marks
    Functions - 2 marks [U] [2]

14. Scanners can be classified into several types, including flatbed, sheet-fed, hand-held etc.
    Scanner definition and types - 1 mark
    Explanation of any 2 types of scanners - 4 marks [U] [2]

15. Memory latency and its components- 1 mark
    DDR4 latency characteristics and memory management - 2 marks
    DDR5 latency characteristics and memory management - 2 marks [An] [3]

16. AR blends digital content with the real world, while VR replaces the real world with a virtual environment.
    Any 5 differences - 1 mark each [An] [3]

(4 x 5 = 20)

Part C
Essay Questions
Answer 2 questions. Each question carries 10 marks

17. Characteristics of fifth generation computers include:
    - Emphasis on artificial intelligence (AI) and natural language processing.
    - Parallel processing and quantum computing are significant advancements.
    - Introduction of superconductors and advanced microprocessor technology.

    These computers aim for better decision-making and human-like thinking capabilities.
    Key characteristics - 5 marks
    Key hardware and software technologies - 4 marks
    Any example - 1 mark [U] [1]

18. Software is a set of instructions or programs that enables a computer to perform specific tasks or functions. System Software manages and controls the hardware, allowing other software to run. It includes operating systems (e.g., Windows, macOS) and utility software (e.g., antivirus programs, file management tools). Application Software are designed for end-users to perform specific tasks, such as word processing, gaming, or browsing the internet. Egs include Microsoft Word, Google Chrome, and Adobe Photoshop.
    Software definition and functions - 2 marks
    System software definition, features/functions and example - 4 marks
    Application software definition, features/functions and example - 4 marks [U] [1]

19. PCI - Peripheral Component Interconnect.PCI is a standard used for connecting internal hardware devices to a computer's motherboard. It was first introduced by Intel in 1992 and played a significant role in expanding the functionality of computers by allowing the attachment of various add-on devices such as sound cards, network cards, modems, and graphic cards.
    Key features - 2 marks
    Advantages - 1 mark
    Disadvantages - 1 mark
    PCIe - Peripheral Component Interconnect Express. PCIe (Peripheral Component Interconnect Express) is a high-speed interface standard used to connect internal hardware components to a computer's motherboard. It is the modern successor to the older PCI and AGP standards, providing much higher performance, flexibility, and scalability. PCIe is widely used today in modern systems for connecting devices such as graphics cards, SSDs, network cards, and other high-performance components.
    Key features - 2 marks
    Advantages - 1 mark
    Disadvantages - 1 mark [U] [2]

20. The three levels of cache—work together to minimize data retrieval delays, improving multitasking performance in multi-core processors
    - L1 (fastest, smallest, core-specific),
    - L2 (larger, slower, shared by a few cores)
    - L3 (largest, slowest, shared among all cores)

    Definition of each level of cache - 1 mark
    Characteristics and impact of each on multicore processors - 3 marks each [An] [3]

(2 x 10 = 20)